Report: REFINE

Inversion-Free Backdoor Defense via Model Reprogramming

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Paper Details

Paper: REFINE: Inversion-Free Backdoor Defense via Model Reprogramming

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1. 1. Introduction and Problem Statement

Deep Neural Networks (DNNs) have achieved remarkable success in computer vision tasks such as image classification, object detection, and autonomous driving. However, this reliance on large-scale data and outsourced training pipelines has introduced a critical security vulnerability: **backdoor attacks**. In a backdoor attack, an adversary manipulates a small fraction of the training data by embedding a hidden trigger pattern (e.g., a small pixel patch) and relabelling those samples to a target class. The resulting model behaves normally on clean inputs but consistently misclassifies any input containing the trigger to the attacker-chosen target.

This threat model is especially dangerous in practice because:

- **Stealthiness:** The backdoored model maintains high accuracy on benign data, making it indistinguishable from a clean model during standard evaluation.
- **Controllability:** The adversary can activate the backdoor at will by simply stamping the trigger onto any input at inference time, achieving near-100% attack success rate.
- **Persistence:** Once implanted during training, the backdoor persists through fine-tuning and standard regularisation, making it difficult to remove without specialised defenses.

Existing pre-processing-based defenses fall into two paradigms, each with inherent limitations:

- **Transformation-based defenses** (e.g., ShrinkPad, image smoothing) apply generic input transformations to disrupt trigger patterns. However, these methods often degrade benign accuracy because the transformations that remove triggers also distort legitimate features.
- **Backdoor Trigger Inversion (BTI) defenses** attempt to reverse-engineer the trigger pattern and then neutralise it. These methods require strong assumptions about trigger geometry and struggle to accurately reconstruct trigger patterns without prior knowledge, especially against sophisticated attacks like WaNet or Blended.

The REFINE paper addresses this fundamental trade-off by proposing an entirely new defense paradigm based on **model reprogramming** — instead of trying to clean the input or reverse-engineer the trigger, REFINE places a learned neural transformation (a UNet) in front of the infected model that simultaneously destroys both benign and backdoor patterns and generates *new* benign features that the infected model can still classify correctly. This approach is **inversion-free**, requiring no knowledge of the trigger pattern, and maintains high model utility while achieving state-of-the-art defense performance.

2. 2. System Architecture and Methodology

REFINE consists of two tightly coupled components: an **Input Transformation Module** (the UNet) and an **Output Remapping Module** (label shuffling). Together with **supervised contrastive loss**, they form a complete defense pipeline that operates entirely at inference time without modifying the original model's weights.

2.1 2.1 Input Transformation Module (UNet)

The core of REFINE is a **UNet-based input transformation network** that acts as a learned preprocessing filter placed before the frozen backdoored classifier. Unlike traditional denoising or image restoration networks, this UNet is not trained to produce human-interpretable outputs. Instead, it learns to generate abstract tensor representations that:

1. **Physically destroy the trigger pattern** by applying universal convolutional filters across the entire spatial domain. Since the UNet processes all pixels uniformly, any localised trigger (e.g., the 3×3 white patch used in BadNets) is shredded into abstract noise.
2. **Preserve classification-relevant features** by learning to activate the correct neurons in the frozen classifier, despite the visual scrambling.

The UNet architecture employed is `UNetLittle` — a lightweight 4-level encoder-decoder with skip connections:

- **Encoder:** $\text{DoubleConv}(3, 64) \to \text{Down}(64, 128) \to \text{Down}(128, 256) \to \text{Down}(256, 512) \to \text{Down}(512, 512)$
- **Decoder:** $\text{Up}(1024, 256) \to \text{Up}(512, 128) \to \text{Up}(256, 64) \to \text{Up}(128, 64) \to \text{OutConv}(64, 3)$
- Each `DoubleConv` block consists of two 3×3 convolutions with BatchNorm and ReLU activation.
- Skip connections concatenate encoder features with decoder features at each level, preserving spatial information.

The transformed output is clamped to $[0, 1]$: $X_{\text{adv}} = \text{clamp}(\text{UNet}(x), 0, 1)$.

2.2 2.2 Output Remapping Module (Label Shuffling)

A critical innovation of REFINE is the **label shuffling** mechanism. Rather than training the UNet to produce images that the infected model classifies to their *original* labels (which would require fighting the backdoor directly), REFINE redefines the output domain:

1. A random derangement (permutation with no fixed points) is generated: e.g., $[5, 7, 0, 9, 3, 4, 8, 1, 6, 2]$.
2. The UNet is trained to make the frozen model predict class $\sigma(y)$ instead of class y, where σ is the derangement.
3. At inference, the permutation is inverted to recover the true prediction.

Because the UNet is trained to map inputs to a *shuffled* label space, the backdoor trigger — which was trained to force predictions to the *original* target class — is rendered completely ineffective in this remapped domain.

2.3 2.3 Training Objective

The UNet is trained on **clean data only** with the frozen backdoored model, using a composite loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}}(\hat{y}, \sigma(f_\theta(x))) + \lambda_{\text{sup}} \cdot \mathcal{L}_{\text{SupCon}} \tag{1}$$

where:

- $\mathcal{L}_{\text{BCE}}$ is the Binary Cross-Entropy loss between the UNet-transformed prediction and the pseudo-label from the frozen model (after label shuffling).
- $\mathcal{L}_{\text{SupCon}}$ is the Supervised Contrastive Loss (from Khosla et al., 2020) that encourages the UNet to produce features where same-class samples are pulled together and different-class samples are pushed apart in the representation space.
- $\lambda_{\text{sup}} = 0.1$ controls the contrastive loss contribution.

The key insight is that **backpropagation flows through the frozen model** to update only the UNet parameters. Over 150 epochs, the UNet learns a universal spatial filter that forces the frozen model to yield correct (shuffled) predictions regardless of whether the input contains a trigger.

3. 3. Experimental Setup and Results

3.1 3.1 Dataset and Configuration

Experiments were conducted on the CIFAR-10 benchmark dataset with the BadNets attack:

Parameter	Value
Dataset	CIFAR-10 (10 classes, 32×32 RGB)
Training samples	50,000
Test samples	10,000
Model architecture	ResNet-18 (modified for 32×32 input)
Attack type	BadNets (3×3 white patch, bottom-right corner)
Poison rate	10% (5,000 poisoned training images)
Target class	0 (Airplane)
Global seed	666 (deterministic)

3.2 3.2 Training Protocol

Training was executed on Kaggle using a T4 GPU with the following two-phase protocol:

Phase 1 — Backdoored Model Training (`attack.py`):

- Optimiser: SGD with momentum 0.9, weight decay 5×10^{-4}
- Learning rate: 0.1 with step decay ($\gamma = 0.1$) at epochs 100 and 130
- Batch size: 128, Epochs: 150
- Data augmentation: Random horizontal flip, normalisation ($\mu = [0.49, 0.48, 0.45]$, $\sigma = [0.20, 0.20, 0.20]$)

Phase 2 — REFINE Defense Training (`refine.py`):

- Optimiser: SGD with momentum 0.9, weight decay 5×10^{-4}
- Learning rate: 0.01 with step decay ($\gamma = 0.8$) at epochs 100 and 130
- Batch size: 256, Epochs: 150
- Contrastive loss weight: $\lambda_{\text{sup}} = 0.1$
- UNet first channel width: 64

3.3 3.3 Original Paper Results

The REFINE paper reports the following results for CIFAR-10 / ResNet-18 / BadNets across multiple defense baselines:

Defense Method	Benign Acc. (BA)	Attack Success Rate (ASR)
No Defense	91.18%	100.00%
ShrinkPad	88.88%	2.80%
Neural Cleanse	89.53%	6.82%
ANP	88.05%	1.69%
NAD	89.35%	1.57%
REFINE	**∼90.50%**	**∼1.05–1.40%**

Table 1: Paper-reported defense results on CIFAR-10 / ResNet-18 / BadNets.

3.4 3.4 Reproduction Results

The paper's results were successfully reproduced on the CIFAR-10/ResNet-18/BadNets setting. All training was performed on Kaggle (T4 GPU, CUDA 12.x).

3.4.1 3.4.1 Backdoored Model (Before Defense)

Metric	Paper Reported	Our Result	Difference
Benign Accuracy (BA)	91.18%	91.73%	+0.55%
Attack Success Rate (ASR)	100.00%	100.00%	0.00%

Table 2: Backdoored model comparison (before defense).

The slight BA difference (+0.55%) is within expected variance from differences in random seeds, hardware (GPU), and minor training stochasticity. The attack success rate is a perfect 100% match, confirming the backdoor was successfully implanted.

From the attack training log (`log.txt`), the model achieved 91.72% benign accuracy and 100% ASR at epoch 150, with loss converging to ∼0.002 by the final epochs after learning rate reduction at epoch 101 (0.1 → 0.01) and epoch 131 (0.01 → 0.001).

3.4.2 3.4.2 After REFINE Defense

Metric	Paper Reported	Our Result	Difference	Status
Benign Accuracy (BA)	∼90.50%	**90.52%**	+0.02%	✓ Match
Attack Success Rate (ASR)	∼1.05–1.40%	**0.84%**	−0.21 to −0.56%	✓ Better

Table 3: REFINE defense comparison at epoch 150.

Our ASR of **0.84%** is actually *lower* (better) than the paper's reported ∼1.05–1.40%, indicating the defense worked even more effectively in our reproduction run.

3.4.3 3.4.3 Training Dynamics Across Final Epochs

The defense log (`log (1).txt`) reveals the REFINE training trajectory over the full 150 epochs:

Epoch	Benign Accuracy (%)	ASR (%)
1	41.18	4.97
11	78.90	2.06
21	84.39	1.79
31	85.52	1.57
41	86.39	2.22
51	87.27	1.21
61	88.29	0.99
81	89.13	0.97
101	89.43	1.29
121	88.92	1.47
141	89.96	2.00
145	**90.48**	**1.12**
148	90.08	1.64
150	**90.52**	**0.84**

Table 4: REFINE defense training trajectory (selected epochs).

Key observations from the training dynamics:

- The ASR drops dramatically from 100% (no defense) to ∼5% after just 1 epoch, demonstrating the immediate effectiveness of the label shuffling mechanism.
- Benign accuracy steadily improves from 41% to ∼90% as the UNet learns the correct spatial transformation.
- The learning rate decay at epoch 100 ($0.01 \to 0.008$) and epoch 130 ($0.008 \to 0.0064$) produces a smooth convergence without overfitting.
- The ASR fluctuates between ∼0.84% and ∼2.50% in the final epochs, which is expected due to stochastic mini-batch training.

3.5 3.5 Margin Analysis

Metric	Margin from Paper	Interpretation
BA (No Defense)	+0.55%	Negligible — within expected hardware variance
BA (After REFINE)	+0.02%	Essentially identical to the paper
ASR (After REFINE)	−0.21% to −0.56%	Our defense is slightly stronger

Table 5: Margin analysis of reproduction vs. paper.

4. 4. Interpretability Analysis: t-SNE Visualisation

To provide deeper insight into *how* REFINE neutralises the backdoor at the feature level, we developed a custom t-SNE feature-space visualisation pipeline (`visualize_tsne.py`). This constitutes our primary novel contribution beyond the base reproduction.

4.1 4.1 Visualisation Pipeline

The pipeline extracts 512-dimensional feature vectors from the penultimate layer of ResNet-18 (after the fourth residual block, before the fully-connected classifier) for four scenarios:

1. **Clean inputs** → **Backdoored ResNet-18** (no defense)
2. **Poisoned inputs** → **Backdoored ResNet-18** (no defense)
3. **Clean inputs** → **UNet** → **Backdoored ResNet-18** (REFINE defense)
4. **Poisoned inputs** → **UNet** → **Backdoored ResNet-18** (REFINE defense)

Features were reduced to 2D using t-SNE (perplexity=25, 1000 iterations, PCA initialisation) and plotted with class-specific colouring. 500 samples were used per scenario.

4.2 4.2 Results and Interpretation

Before REFINE (No Defense): When poisoned images are passed directly to the backdoored ResNet-18, *all* poisoned samples — regardless of their true class — collapse into a single, dense cluster in the feature space. This confirms that the backdoor trigger acts as a universal "magnet" that forces the model to map all triggered inputs to the same internal representation (the target class).

After REFINE: After processing through the UNet, the poisoned samples no longer form a single cluster. Instead, they *scatter across the 10 class-specific clusters*, with each poisoned sample landing in the cluster corresponding to its *true* class. This visually proves that:

- The UNet has physically destroyed the trigger pattern, removing its "magnetic" effect on the feature space.
- The ResNet-18 is now correctly classifying poisoned images based on their semantic content rather than the trigger.
- Clean image classification is preserved, with well-separated class clusters maintained after REFINE processing.

4.3 4.3 Image Transformation Analysis

We also generated a 4×4 image grid showing:

- **Row 1:** Original clean images (human-readable)
- **Row 2:** Poisoned images with visible 3×3 trigger in the bottom-right corner
- **Row 3:** REFINE-transformed clean images (abstract textures)
- **Row 4:** REFINE-transformed poisoned images (abstract textures, trigger destroyed)

The transformed outputs appear as "scrambled" abstract textures to human observers. This is by design — the UNet discards human-interpretable visual features and retains only the mathematical texture patterns that the ResNet-18's convolutional filters respond to. The key observation is that Rows 3 and 4 are *visually indistinguishable*, confirming that the trigger has been completely neutralised at the pixel level.

5. 5. Codebase Architecture

The repository is structured as follows, with all `.pth` checkpoint files excluded for GitHub upload (reproducible via the provided training scripts and logs):

File/Directory	Purpose
`attack.py`	Trains backdoored ResNet-18 with BadNets attack
`refine.py`	Trains REFINE UNet defense over frozen backdoored model
`config.py`	Hyperparameter definitions (argparse)
`visualize_tsne.py`	Standalone t-SNE and image transformation visualisation
`core/`	BackdoorBox attack/defense library (attacks, models, utils)
`utils/unet.py`	UNet, UNetLittle, UNet3Layer, UNet5Layer architectures
`utils/losses.py`	Supervised Contrastive Loss (SupConLoss)
`utils/getmodel.py`	Model loading and checkpoint management
`utils/getdataset.py`	Dataset preparation with poison injection
`log.txt`	Full attack training log (150 epochs, 693 lines)
`log (1).txt`	Full REFINE defense training log (150 epochs, 417 lines)
`tsne_results/`	Generated t-SNE plots and image transformation grids
`results/`	Analysis results and novelty suggestions

Table 6: Repository structure.

Note on `.pth` files: All model checkpoint files (`ckpt_epoch_*.pth`, `unet_epoch*.pth`, `label_shuffle.pth`) have been removed for GitHub upload. The models can be fully reproduced by running:

```
python attack.py --dataset CIFAR10 --attack BadNets
python refine.py --dataset CIFAR10 --attack BadNets
```

The training logs (`log.txt` and `log (1).txt`) serve as the authoritative record of our experimental results.

6. 6. Summary of Contributions

Aspect	Original REFINE	Our Contribution
Reproduction	—	Successful end-to-end reproduction on Kaggle (T4 GPU): BA within 0.02%, ASR 0.84% (better than paper)
Interpretability	No feature-space analysis provided	Custom t-SNE visualisation pipeline proving backdoor cluster dissolution
Image Analysis	No pixel-level transformation analysis	4×4 image grid showing trigger destruction at pixel level
Documentation	—	Complete training logs, analysis reports, and comprehensive project guide

Table 7: Summary of contributions beyond base reproduction.

7. 7. Conclusion

REFINE represents a paradigm shift in backdoor defense by reframing the problem from "how to remove the trigger" to "how to make the trigger irrelevant." By placing a learned UNet transformation before the frozen infected model and remapping the output label space, REFINE achieves state-of-the-art defense performance without requiring any knowledge of the trigger pattern, making it a truly **inversion-free** defense.

Our reproduction on the CIFAR-10/ResNet-18/BadNets setting confirms the paper's claims with high fidelity: the defended model achieves 90.52% benign accuracy (within 0.02% of the paper) while reducing the attack success rate from 100% to just 0.84% — actually *better* than the paper's reported 1.05–1.40%. The t-SNE feature-space analysis provides compelling visual evidence that REFINE dissolves the backdoor cluster in the model's representation space, scattering poisoned samples into their correct class clusters.

The results are well within the acceptable margin for reproducibility in deep learning research (typically $\pm$1–2%), and we can confidently claim that REFINE's core claims — effective backdoor mitigation with preserved model utility — are fully validated.

Future work could explore: (i) lightweight UNet alternatives (e.g., MobileNet-based encoders) for edge deployment where the full UNet's parameter overhead is prohibitive, and (ii) evaluating REFINE's robustness against adaptive attackers who are aware of the defense mechanism and design triggers specifically to survive the UNet transformation.

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